

Business & Data Analytics

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DATA CLEANING

Overview: Data Cleaning

Full definition:

Data Cleaning is identifying and correcting errors, inconsistencies and anomalies in a dataset to ensure that the data is accurate, reliable and suitable for analysis

Removing inconsistencies and preparing the data for analysis



Common Issues:



Missing values
Inaccurate entries
Duplicate data



Why we need to clean the data Btw?

Why we Clean the Data First?



Data Quality and Integrity

Uncleaned data (incomplete, inconsistent and inaccurate)
can result incorrect conclusion

Affecting the validity of the entire analytic process

Why we Clean the Data First?



Real-world data is not always clean (but why?)

Real-world data is seldom clean. It is often with missing values, duplicates, outliers and inconsistencies.

Outliers- an observation that lies an abnormal distance from other values in a random sample from a population

Why we Clean the Data First?



Identification of Errors

Identifying errors and solving them is a good foundation to the data analytic process.

Why we Clean the Data First?



Enhancing Productivity

To significantly enhance the productivity of subsequent tasks. It minimizes unnecessary additional tasks like revisiting analyses.

Why we Clean the Data First?



Ethical and Regulatory Compliance

Many organizations are subject to data privacy and regulatory requirements that mandate the removal of sensitive or personally identifiable information

COMMON DATA ISSUES



1. Incorrect Values

COMMON DATA ISSUES



2. Inconsistent Data Format

COMMON DATA ISSUES



3. Outliers

COMMON DATA ISSUES



4. High Cardinality

COMMON DATA ISSUES



Set or group of things has a lot of unique elements.

**Example: names – high cardinality
Gender – low cardinality**

COMMON DATA ISSUES



5. Duplicate Records

COMMON DATA ISSUES



6. Highly correlated variables ?

Two or more variables in a dataset are strongly related to each other. When one variable changes, the other variables tends to change in a predictable

COMMON DATA ISSUES



Give examples.

Why we Clean the Data First?



- 1. Incorrect Values**
- 2. Inconsistent Data Format**
- 3. Outliers**
- 4. High Cardinality**
- 5. Duplicate Records**
- 6. Highly correlated variables**

Data Issues : Incorrect Values



Data Quality and Integrity

Gender	Frequency
Male	8523
Female	7549
.	673

What can you say about the dataset?

Data Issues : Incorrect Values



Set Values to Prevent Incorrect Values:

1. Minimum- maximum
2. Criteria
3. Positive/Negative
4. Values only
5. Non-zero criteria

Data Issues : Incorrect Values



Domain expertise and knowledge of data sources are needed to determine whether a values is correct

How to prevent missing values especially if you are system developer or IT specialist?

Data Issues : Incorrect Values



Treatment:

- Remove the observation
- Treat as missing
- Replace with correct value
- Identify root cause and correct from the source

Data Issues: Inconsistent Data Format



- Inconsistent format within a single column
- Inconsistent data type of a variable coming from different data sources or timeline
- Typical problematic variables:
 - Date (formats)
 - Zip codes (numeric / character)
 - Telephone number

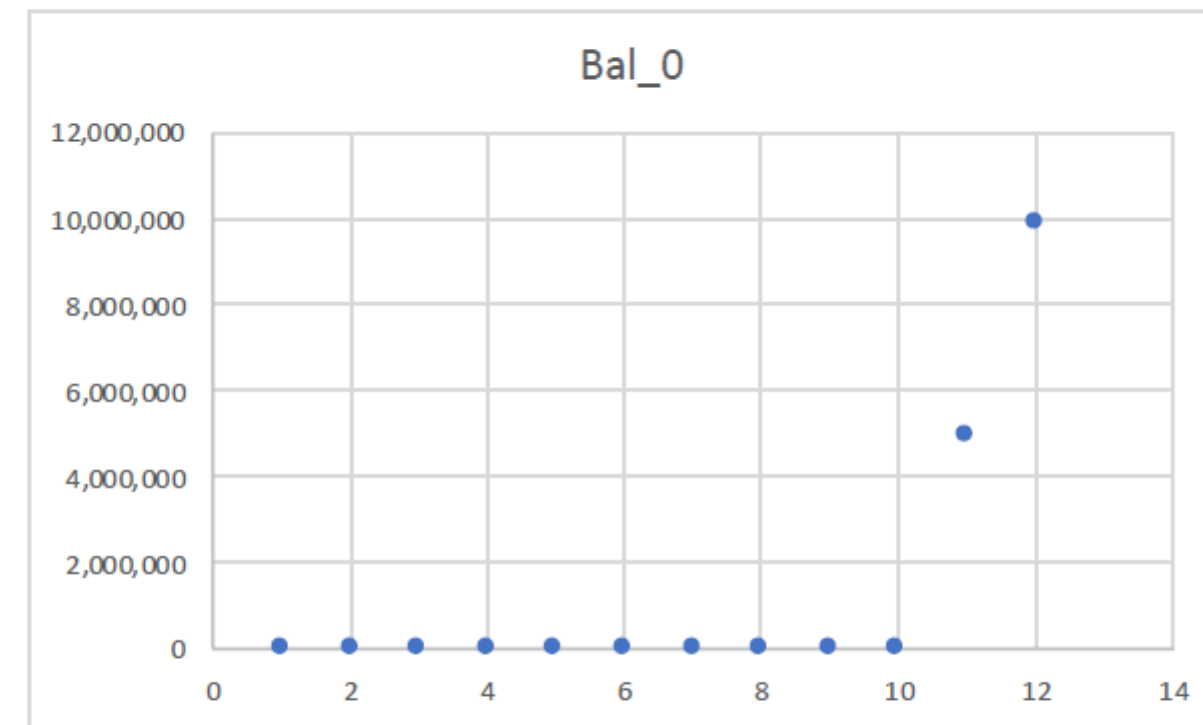
Data Issues: Outliers

Outliers

-> For numeric variables

What : An observation that diverges from an overall pattern on a sample

Effect : If not diagnosed correctly, may mis-lead the modeling process and result to inaccurate and biased model



Data Issues: Outliers

Causes of Outliers:

- Data entry errors
 - Measurement errors
 - Experimental errors
 - Intentional errors
 - Data processing errors
 - Natural occurrence
- 
- A decorative graphic consisting of several thin, light blue wavy lines that flow from the right side of the slide towards the center, creating a sense of movement and depth.

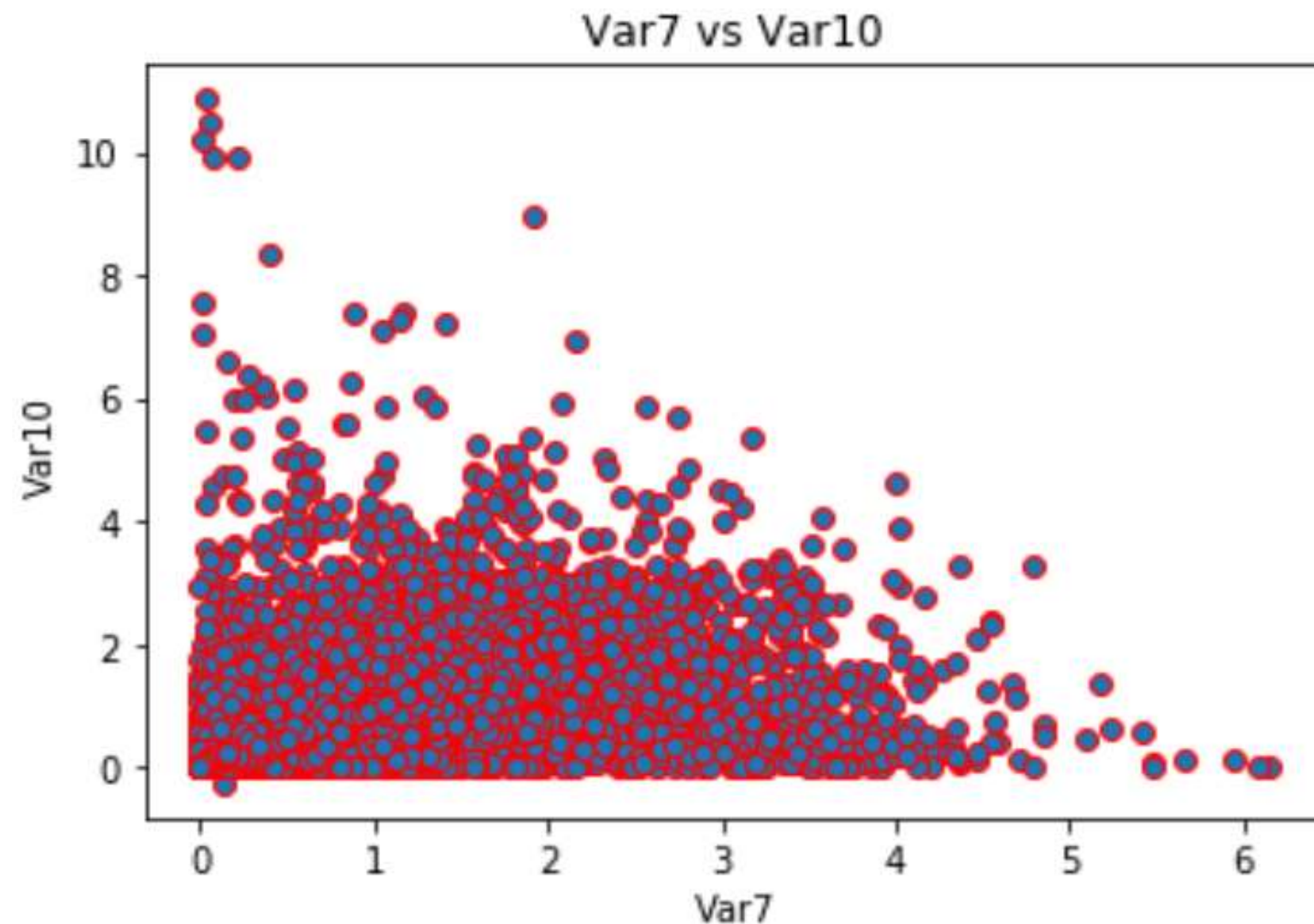


How to detect Outliers?

Data Issues: Outliers

Visualization

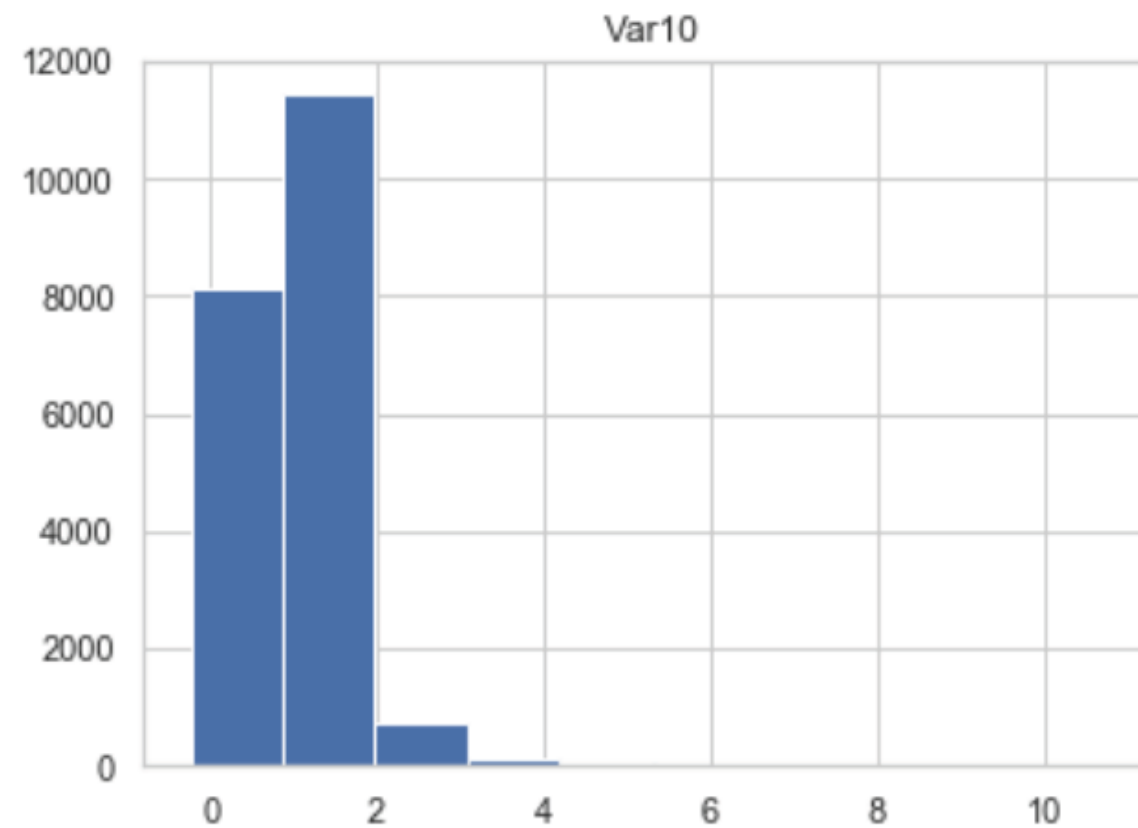
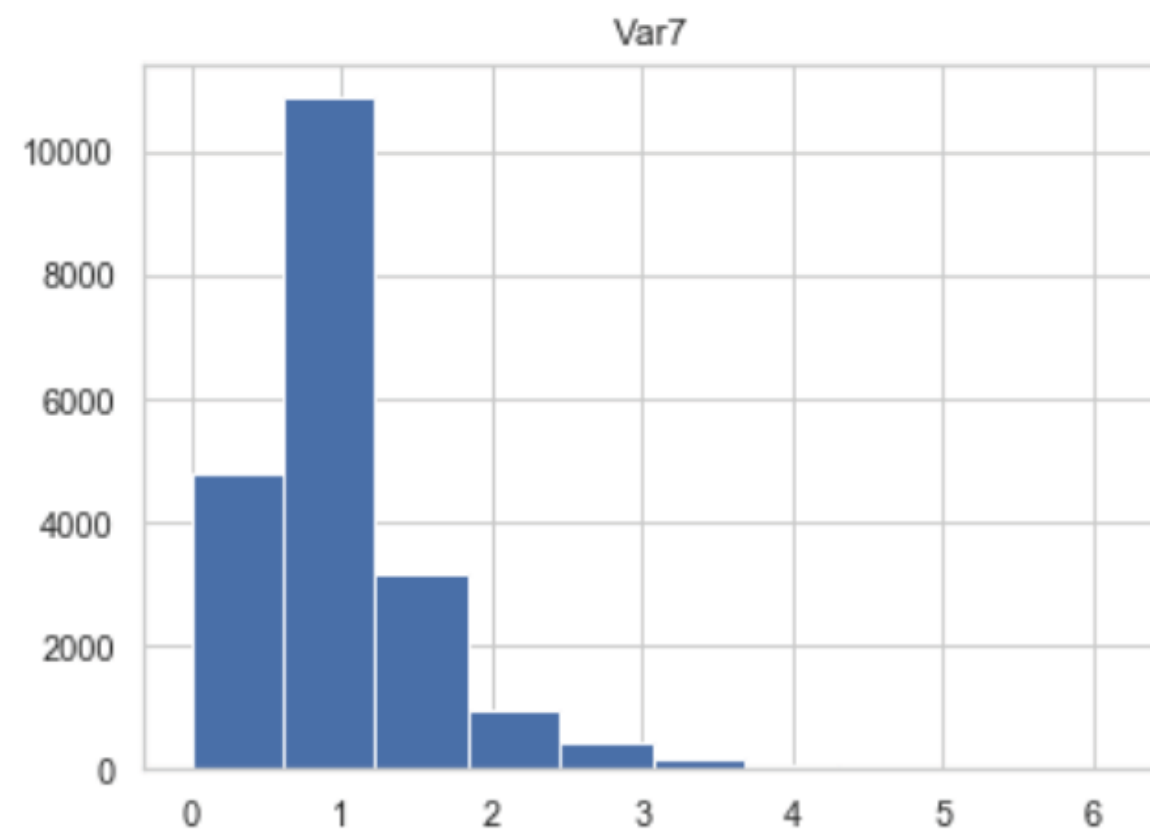
- Scatter Plot – 2 variables plotted on x & y axis



Data Issues: Outliers

Visualization

- Histogram



The histogram gives an idea both of the presence of outlier and the distribution of the dataset

Data Issues: Outliers

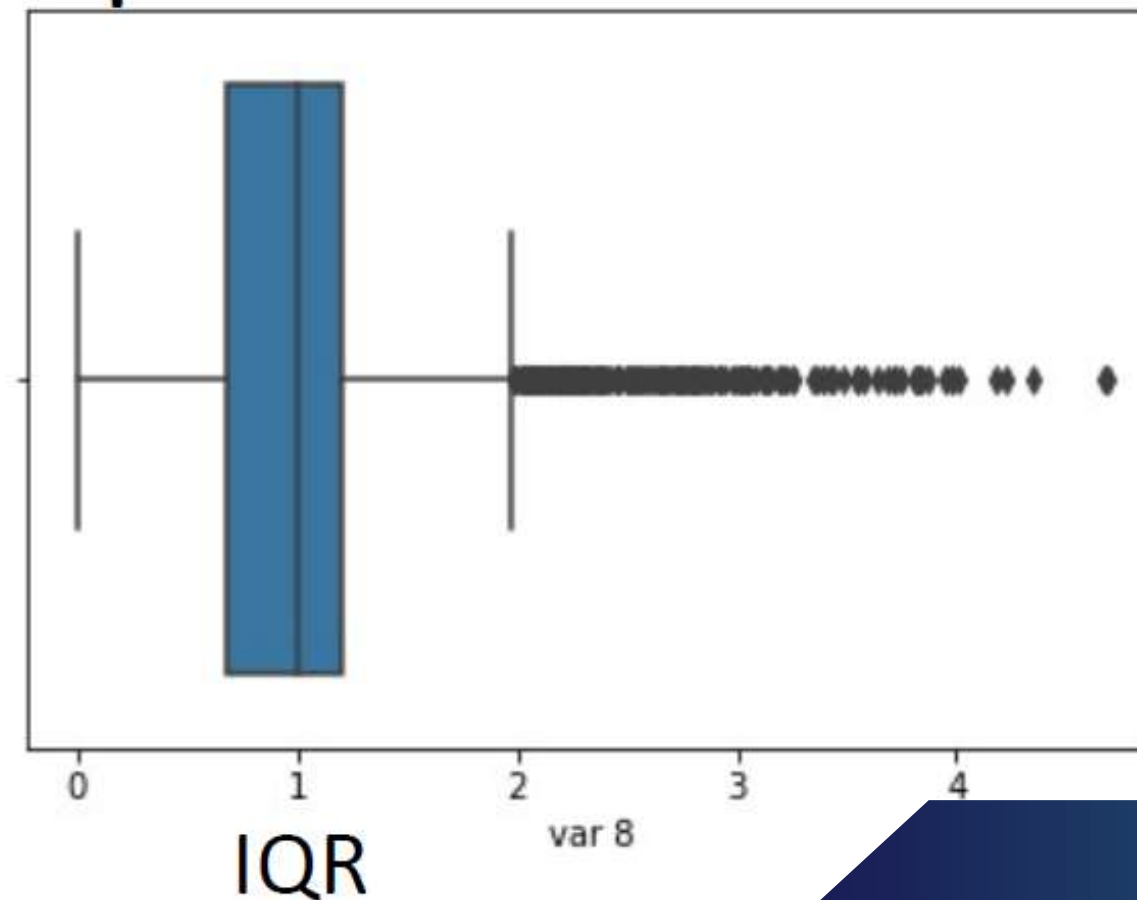
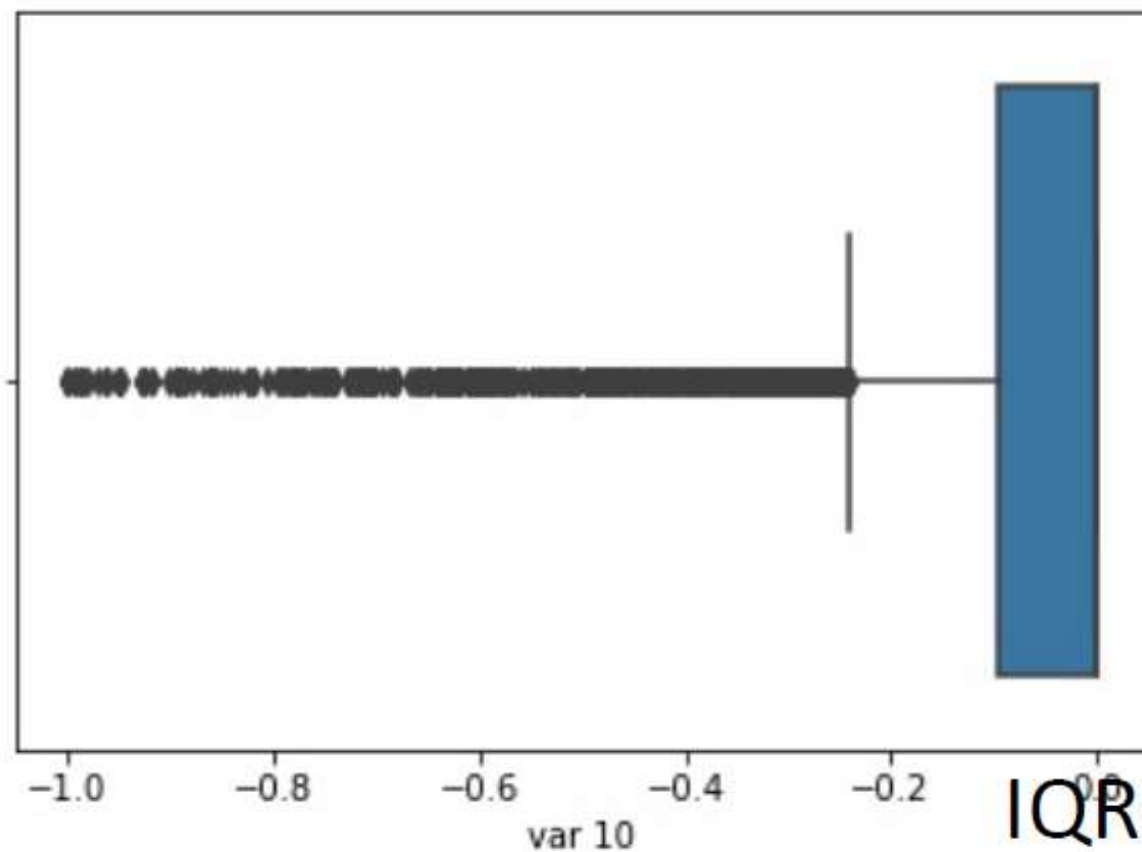
Visualization

- Box plot – is a standardized way of displaying the distribution of data based on the five number summary: minimum, first quartile, median, third quartile, and maximum

Data Issues: Outliers

Box-Plot (box & whisker plot)

- **Outliers** are either $3 \times IQR$ or more above the third quartile or $3 \times IQR$ or more below the first quartile.



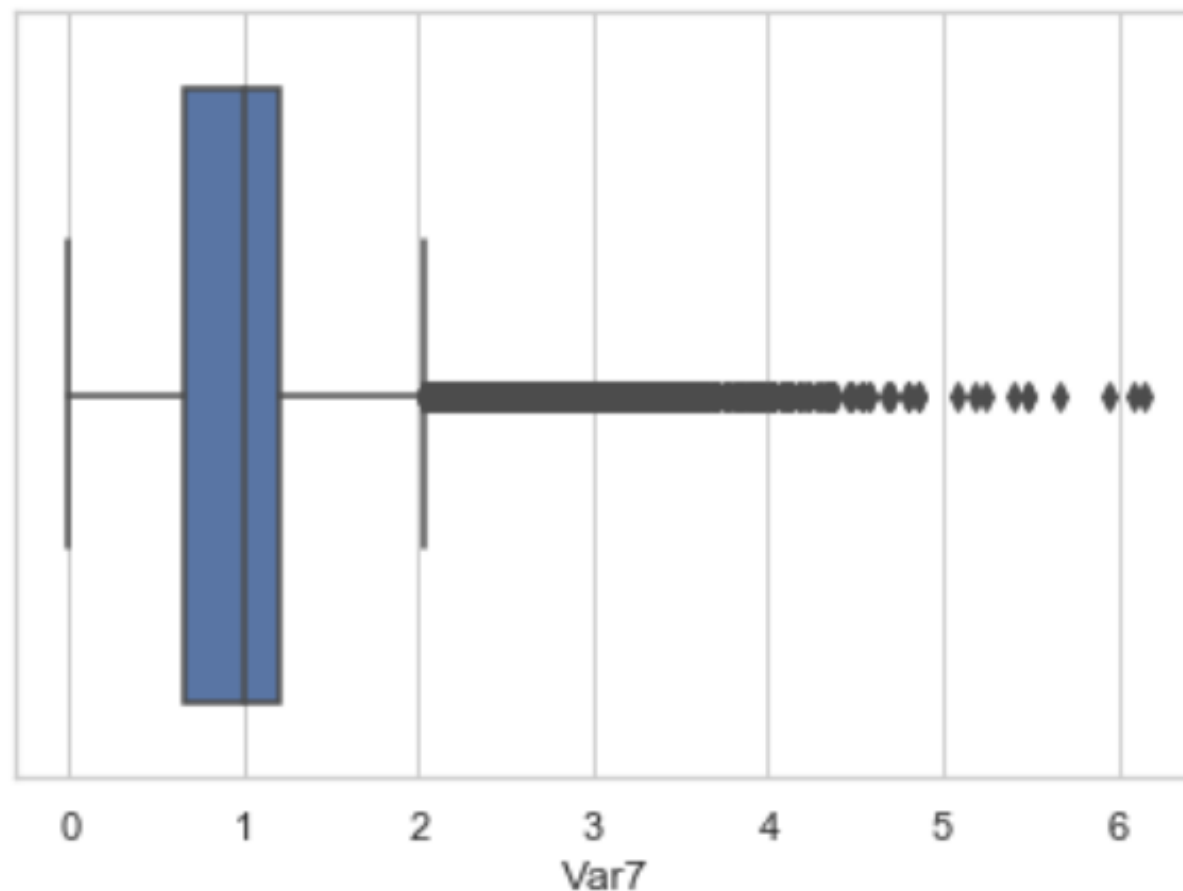
Data Issues: Outliers

Mathematical Function:

- Inter-Quartile Range

Outliers : Values below $Q_1 - 2.5(Q_3 - Q_1)$

Values Above $Q_3 + 2.5(Q_3 - Q_1)$



Adjust the multiplier(2.5) if the resulting observations are too many

Data Issues: Outliers

Interquartile Range Result: Var 7

IQR : Var7 0.555398

dtype: float64

Upper : Var7 2.593521

dtype: float64

Lower : Var7 -0.738867

dtype: float64

19848	yes
19871	yes
19882	yes
19887	yes
20008	yes
20023	yes
20037	yes
20049	yes
20170	yes
20173	yes
20198	yes
20309	yes
20347	yes
20370	yes
20374	yes
20380	yes
20402	yes
20529	yes

[587 rows x 1 columns]

Data Issues: Outliers

Handling of Outlier:

- investigate if this is accurate, reasonable, important
 - if yes :
 - Retain as is
 - Replace with approximate using inference
 - Transform the variable
 - Bin the variable
 - Delete record
 - Create a separate model for the outliers
 - If not :
 - Delete record
 - Replace with approximate using inference

NOTE: An outlier may be what most disturbs your analysis, but may also be exactly what you are looking for

Data Issues: Missing Values

Handling (case to case depending on the objective of the study):

- Impute
 - Mean
 - Assigned value
- Delete the record
- Replace with zeros

NOTE: Missing values could be a powerful part of the model.
Be careful on what handling technique to use

Data Issues: High Cardinality

High cardinality - categorical attributes that contain a very large number of distinct value (example: zip codes)

Handling :

Create wider categories

Data Issues: High Cardinality

Philippine Provincial ZIP Codes

- ⇒ [Abra](#)
- ⇒ [Agusan del Norte & Agusan del Sur](#)
- ⇒ [Aklan](#)
- ⇒ [Albay](#)
- ⇒ [Antique](#)
- ⇒ [Aurora](#)
- ⇒ [Basilan](#)
- ⇒ [Bataan](#)
- ⇒ [Batanes](#)
- ⇒ [Batangas](#)
- ⇒ [Benguet](#)
- ⇒ [Biliran Sub-Province](#)
- ⇒ [Bohol](#)
- ⇒ [Bukidnon](#)
- ⇒ [Bulacan](#)
- ⇒ [Cagayan](#)
- ⇒ [Camarines Norte & Camarines Sur](#)
- ⇒ [Camiguin](#)
- ⇒ [Cavite](#)
- ⇒ [Compostela Valley](#)
- ⇒ [Cebu Province](#)
- ⇒ [Cotabato North & South](#)
- ⇒ [Davao del Norte, del Sur & Oriental](#)
- ⇒ [Guimaras Sub-Province](#)
- ⇒ [Ifugao](#)
- ⇒ [Ilocos Norte & Ilocos Sur](#)
- ⇒ [Iloilo Province](#)
- ⇒ [Isabela](#)
- ⇒ [Kalinga-Apayao](#)
- ⇒ [Laguna](#)
- ⇒ [Lanao del Norte & Lanao del Sur](#)
- ⇒ [La Union](#)
- ⇒ [Leyte](#)
- ⇒ [Maguindanao](#)
- ⇒ [Marinduque](#)
- ⇒ [Masbate](#)
- ⇒ [Mindanao Occidental & Oriental](#)

Philippine
zip Code



Luzon
Visayas
Mindanao
NCR

Data Issues: Duplicate Records

Types

- Duplicate ID but columns are different
- All columns are the same
- ID are unique but all other columns are the same

Handling

- Case to case based on result of investigation

Data Issues: Duplicate Records

Types

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- All columns are the same
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Handling

- Case to case based on result of investigation

Data Cleaning

Can be applied when data are submitted to you

Business & Data Analytics



Your research work will revolve around the topic of Data Transformation. To guide your exploration and understanding, you will be provided with a PowerPoint presentation and a set of guide questions. The primary objective is for you to immerse yourselves in the subject matter and conduct thorough research. The insights gained from this research will be valuable not only for an upcoming quiz but also for the programming tasks in the Business Analytics Laboratory scheduled for October 18, 2023. The submission deadline for your research work is set for October 16, 2023, and you will be able to submit your findings via a Google Form, which will be provided to you at a later date.

This research endeavor aims to empower you with a deeper understanding of data transformation, its methods, and applications. We encourage you to explore beyond the guide questions, delve into real-world scenarios, and document your findings. Your dedicated research will not only contribute to your academic success but also enhance your ability to apply these concepts in practical data analytics.

A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values

Methods

Smoothing: Remove noise from data

Attribute/feature construction

New attributes constructed from the given ones

Aggregation: Summarization, data cube construction

Normalization: Scaled to fall within a smaller, specified range

min-max normalization

z-score normalization

normalization by decimal scaling

Discretization: Concept hierarchy climbing

- Min-max normalization: to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

- Z-score normalization (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

- Ex. Let $\mu = 54,000$, $\sigma = 16,000$. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

- Normalization by decimal scaling

$$v' = \frac{v}{10^j} \quad \text{Where } j \text{ is the smallest integer such that } \text{Max}(|v'|) < 1$$

THANK YOU

